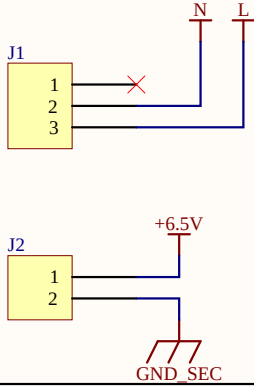
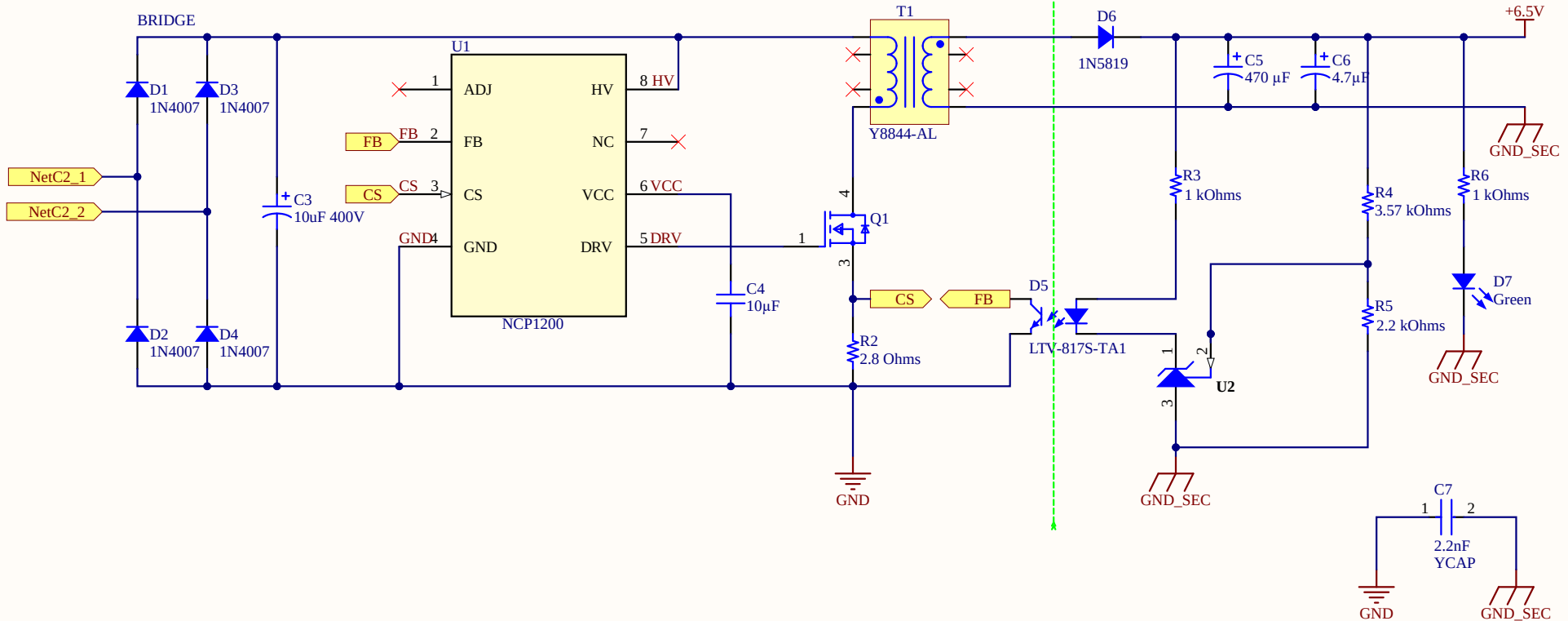
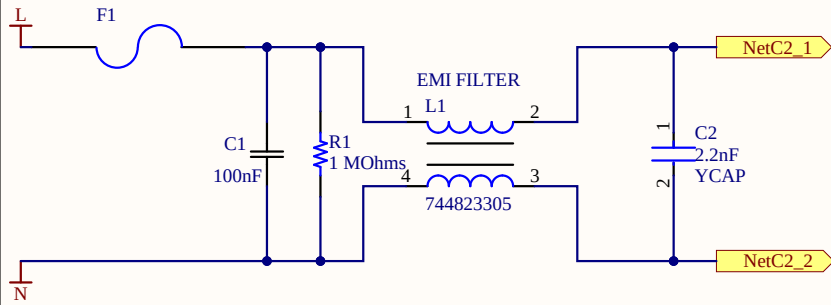


# CONNECTORS

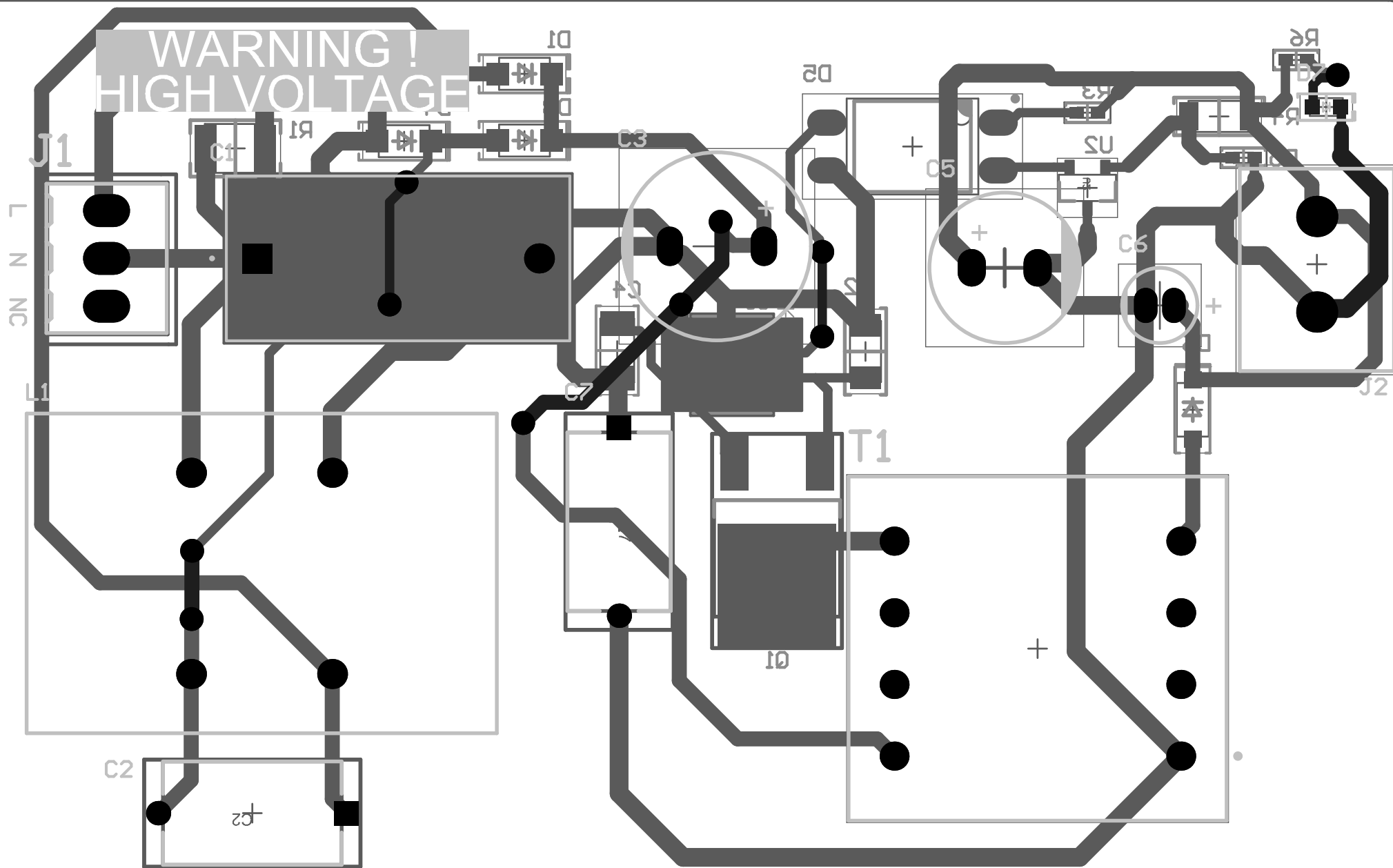


# AC(230V @ 50Hz) INPUT



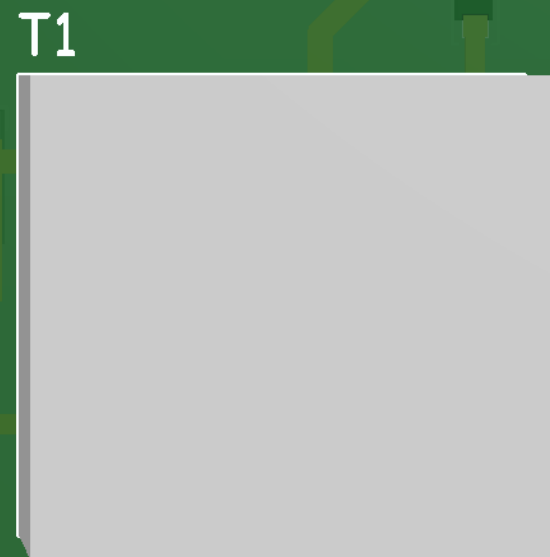
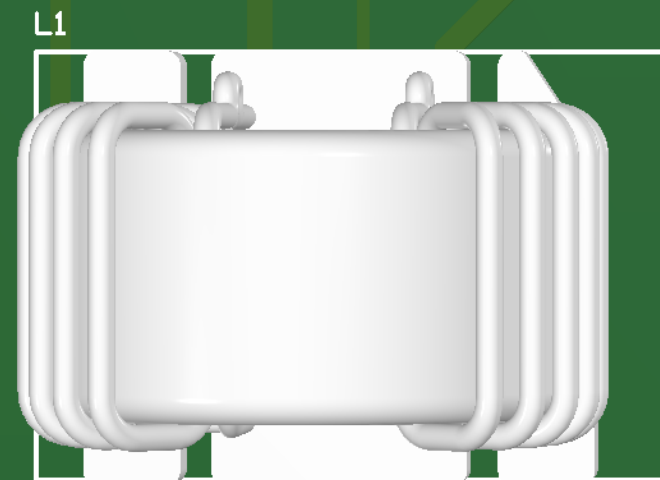
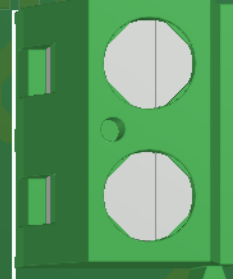
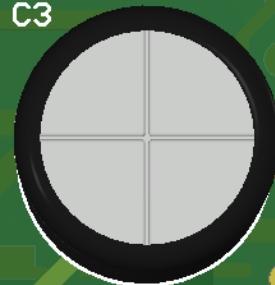
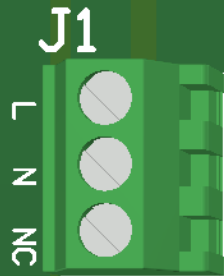
|       |                            |  |              |
|-------|----------------------------|--|--------------|
| Title |                            |  | FLYBACK SMPS |
| Size  | Number                     |  | Revision     |
| A     |                            |  |              |
| Date: | 9/06/2025                  |  | Sheet of     |
| File: | C:\Users\...\Sheet1.SchDoc |  | Drawn By:    |

WARNING!  
HIGH VOLTAGE



SMPS Mains → 6.5V Flyback

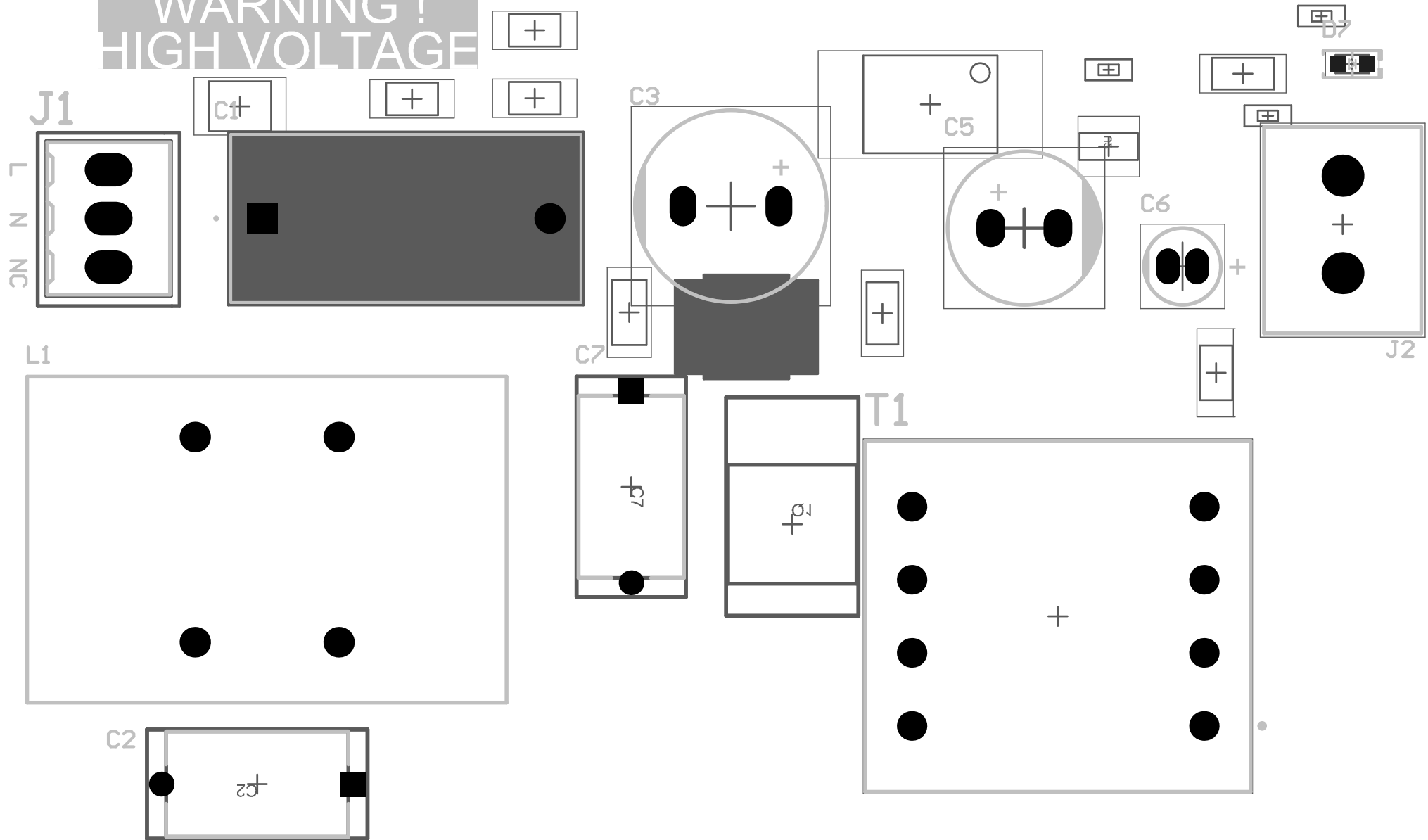
WARNING!  
HIGH VOLTAGE



SMPS Mains → 6.5V Flyback

| Line # | Name              | Description                                                                                                                                | Designator     | Revision ID      | Revision State | Revision Status | Quantity | Manufacturer 1            | Manufacturer Part Number 1 | Manufacturer Lifecycle 1       | Supplier 1        | Supplier Part Number 1       | Supplier Unit Price 1 | Supplier Subtotal 1 |
|--------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------|----------------|-----------------|----------|---------------------------|----------------------------|--------------------------------|-------------------|------------------------------|-----------------------|---------------------|
|        | VY1222M37Y5VQ63V0 | 2200 pF±20%<br>500VAC Ceramic Capacitor Y5V(F) Radial, Disc                                                                                | C2, C7         | CMP-001-00084-1  | Draft          | Up to date      | 2        | Vishay                    | VY1222M37Y5VQ63V0          | Volume Production              | Arrow Electronics | VY1222M37Y5VQ63V0            | 0.1776                | 0.3552              |
|        | TL431AQDB2RQ1     | Automotive Catalog Adjustable Precision Shunt Regulator, 34 ppm / degC, 100 mA, -40 to 125 degC, 3-pin SOT-23 (DBZ), Green (RoHS&no Sb/Br) | U2             |                  | Not managed    |                 | 1        |                           |                            |                                |                   |                              |                       |                     |
|        | 10uF 400V         | CAP 10 UF 20% 400 V                                                                                                                        | C3             |                  | Not managed    |                 | 1        | Würth Elektronik          | 860241375002               | Unknown                        | DigiKey           | 732-9301-1-ND                | 0.88                  | 0.88                |
|        | 4.7µF             | CAP ALUM 4.7UF 20% 16V RADIAL                                                                                                              | C6             |                  | Not managed    |                 | 1        |                           |                            |                                | DigiKey           | 493-10248-1-ND               |                       |                     |
|        | 470 µF            | CAP ALUM 470UF 20% 16V RADIAL                                                                                                              | C5             |                  | Not managed    |                 | 1        | Nichicon                  | UKW1C471MPD1TD             | Volume Production              | DigiKey           | 493-10758-1-ND               | 0.7                   | 0.7                 |
|        | 10µF              | CAP CER 10UF 50V X5R 1206                                                                                                                  | C4             |                  | Not managed    |                 | 1        |                           |                            |                                | DigiKey           | 478-9979-1-ND                |                       |                     |
|        | 2.8 Ohms          | CPF1206 2R8 0.1% 25PPM 1K RL                                                                                                               | R2             |                  | Not managed    |                 | 1        | TE Connectivity / Neohm   | CPF1206B2R8E1              | Volume Production              | DigiKey           | 1712-CPF1206B2R8E1CT-ND      | 0.35                  | 0.35                |
|        | 1N5819            | DIODE SCHOTTKY 40V 1A SOD123                                                                                                               | D6             |                  | Not managed    |                 | 1        | Diodes Inc.               | 1N5819HW-7-F               | Volume Production              | DigiKey           | 1N5819HW-FDICT-ND            | 0.34                  | 0.34                |
|        | Y8844-AL          | Flyback Transformer, 3.5 W, -40 to 85 degC, 18.3 x 20.1 x 14 mm Package                                                                    | T1             |                  | Not managed    |                 | 1        |                           |                            |                                |                   |                              |                       |                     |
|        | 0679H2000-05      | FUSE BOARD MNT 2A 350VAC 72VDC                                                                                                             | F1             |                  | Not managed    |                 | 1        |                           |                            |                                | DigiKey           | 507-2049-1-ND                |                       |                     |
|        | STD1NK60T4        | N-Channel 600V 1A (Tc) 30W (Tc) Surface Mount DPAK                                                                                         | Q1             | CMP-007-000000-1 | Draft          | Up to date      | 1        | STMicroelectronics        | STD1NK60T4                 | Volume Production              | Newark            | 33R1142                      | 0.221                 | 0.221               |
|        | Green             | NEXT GENERATION OF CHIP LED 0603                                                                                                           | D7             |                  | Not managed    |                 | 1        | Osram Opto                | LTQ99G.01-R2T1-25-1-5-R18  | Unknown                        | DigiKey           | 475-LTQ99G.01-R2T1-25-1CT-ND | 0.15                  | 0.15                |
|        | 100nF             | No Description Available                                                                                                                   | C1             |                  | Not managed    |                 | 1        |                           |                            |                                |                   |                              |                       |                     |
|        | NCP1200           | No Description Available                                                                                                                   | U1             |                  | Not managed    |                 | 1        |                           |                            |                                |                   |                              |                       |                     |
|        | LTV-817S-TA1      | OPTOISOLATR 5KV TRANSISTOR 4-SMD                                                                                                           | D5             |                  | Not managed    |                 | 1        | Lite-On                   | LTV-817S-TA1               | Volume Production              | DigiKey           | 160-1893-1-ND                | 0.32                  | 0.32                |
|        | 1 MOhms           | RES 1M OHM 1% 3/4W 1210                                                                                                                    | R1             |                  | Not managed    |                 | 1        | TE Connectivity / AMP     | 6-2176329-1                | Volume Production              | DigiKey           | A130635CT-ND                 | 0.3                   | 0.3                 |
|        | 220 Ohms          | RES 3.57K OHM 1% 1/16W 0402                                                                                                                | R4             |                  | Not managed    |                 | 1        | Yageo Group               | RC0402FR-073K57L           | Volume Production              | DigiKey           | YAG3150CT-ND                 | 0.1                   | 0.1                 |
|        | 1 kOhms           | RES SMD 1K OHM 0.1% 1/16W 0402                                                                                                             | R3, R6         |                  | Not managed    |                 | 2        | TE Connectivity           | CPF0402B1K0E               | Volume Production              | DigiKey           | A119928CT-ND                 | 0.19                  | 0.38                |
|        | 2.2 kOhms         | RES SMD 2.2K OHM 0.5% 1/16W 0402                                                                                                           | R5             |                  | Not managed    |                 | 1        | Susumu                    | FR0510P-222-D              | Not Recommended for New Design | DigiKey           | FR05P2.2KDCT-ND              | 0.1                   | 0.1                 |
|        | 1N4007            | STANDARD RECTIFIER 1000V SOD-123                                                                                                           | D1, D2, D3, D4 |                  | Not managed    |                 | 4        | Sangdest Microelectronics | 1N4007FL                   | Volume Production              | DigiKey           | 1655-1N4007FLCT-ND           | 0.13                  | 0.52                |
|        | 282837-2          | TERM BLK 2P SIDE ENT 5.08MM PCB                                                                                                            | J2             |                  | Not managed    |                 | 1        | TE Connectivity           | 282837-2                   | Volume Production              | DigiKey           | A113320-ND                   | 0.74                  | 0.74                |
|        | Part Number       | TERM BLK 3P SIDE ENT 2.54MM PCB                                                                                                            | J1             |                  | Not managed    |                 | 1        | TE Connectivity           | 282834-3                   | Volume Production              | DigiKey           | A98334-ND                    | 3.93                  | 3.93                |
|        | 744823305         | THT Common Mode Power Line Choke WE-CMB, L = 5.00 mH                                                                                       | L1             |                  | Not managed    |                 | 1        |                           |                            |                                |                   |                              |                       |                     |

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